Markov Chains:
What Comes ***Next***?

As **generative AI/ML** systems are becoming more accessible and pervasive in youths' lives, there is a need to consider **how we can support their understandings of the underlying mechanisms of such technologies.**

AI/ML is about prediction

- **Markov Chains** are foundational for **probabilistic modeling**, particularly relevant in large language models (LLMs)
- Most learning opportunities **assume access to computers, tablets, prior coding knowledge, or familiarity with AI concepts** (e.g., Tseng et al., 2023)

- Our activity adopts an **unplugged approach** to make probabilistic modeling & data-related concepts (Olari & Romeike, 2024) more accessible and easy to implement.

- Through **scaffolded instruction**, the activity gradually guides learners from observing patterns to constructing simple Markov models.

- The design allows the activity to be **plugged into a variety of formal and informal learning settings** (e.g., classrooms & museums).

Markov Chains are a ML method to predict what _____ comes ***next***

word

letter

number

emoji

Markov Chains are a ML method to predict what _____ comes ***next***

word
letter
number
emoji

tokens

Markov chains can be used for

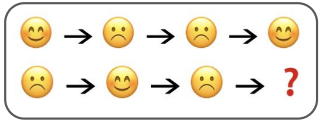
- 🌦️ Weather predictions
- 👾 Games (and gambling)
- 💰 Stock market
- 🕺 Predicting behaviors
- 🧬 Analyzing biological and chemical sequences

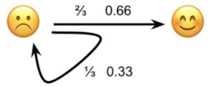
Activity Overview

Activity 1 — What Comes Next?: Introducing Markov Chains

Using simple sequences of face emojis, learners will predict the following face emoji.

Key Concept: Prediction is based on prior data





Activity 2 — N-imals: Introducing N-grams

Using simple sequences of animal emojis, learners will predict the following animal emoji following the last n-number of tokens (e.g., last two for bigrams).

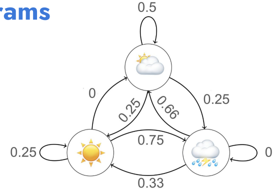
Key Concept: Memory length in predictive systems

Answer Key		
😺🐰 → 😺	1/7	= 0.14
😺🐰 → 🐰	0/7	= 0
😺🐰 → 🐙	4/7	= 0.57
😺🐰 → 🐨	0/7	= 0
😺🐰 → 🐶	2/7	= 0.29

Activity 3 — Sunny, Rainy, Cloudy: Introducing State Diagrams

Using a sequence of weather data, learners will create a state diagram and then use it to predict (most likely) future weather.

Key Concepts: State transition diagrams, prediction models as a system



Activity 4 — Rock, Paper, Scissors: Rapid Prototyping of Datasets

Using the familiar game of Rock, Paper, Scissors, learners will iteratively build their own datasets, resulting in a final round determined by the output of their dataset.

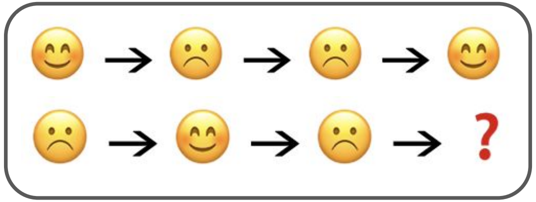
Key Concepts: Data quantity and quality, bias, overfitting

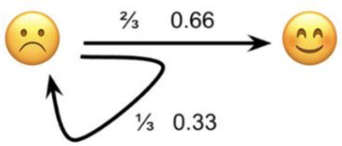
Activity 1. What Comes Next?

Activity 1

What Comes Next?: Introducing Markov Chains

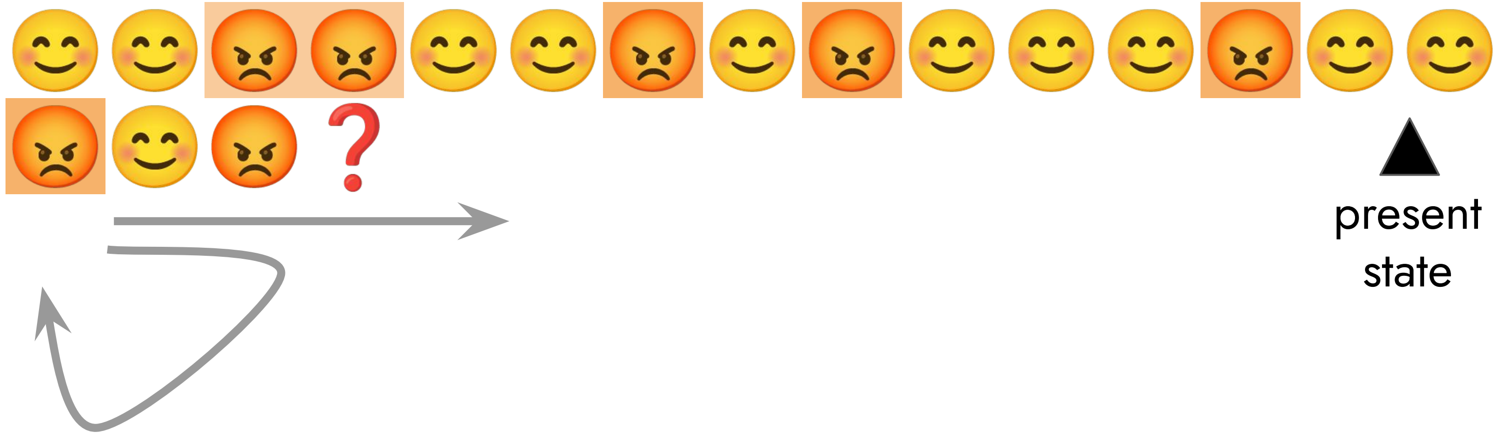
Using simple sequences of face emojis, learners will predict the following face emoji.

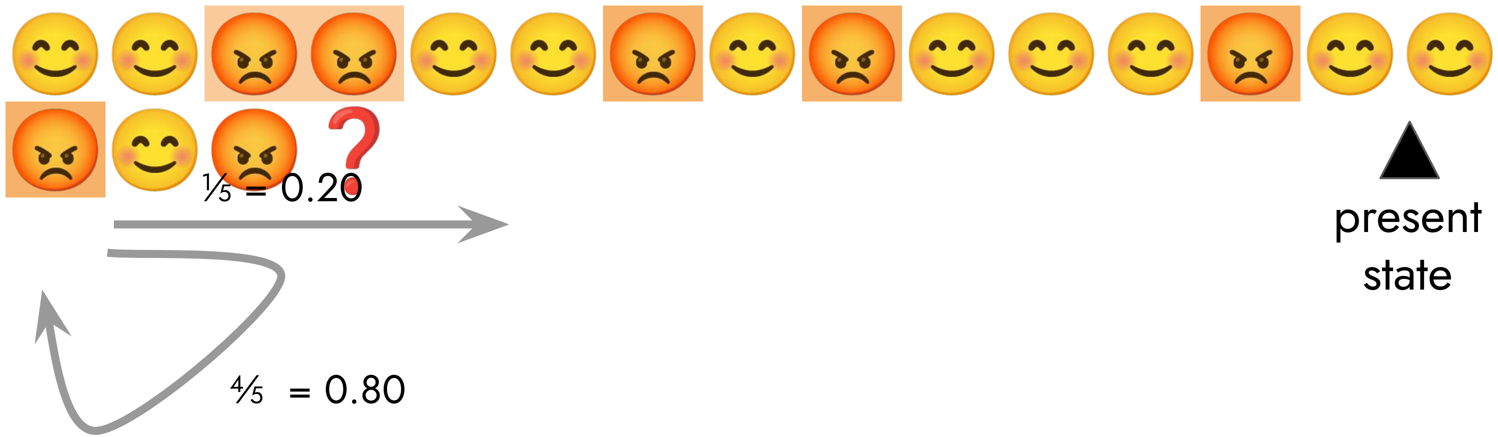




Key Concept: Prediction is based on prior data
Key Terms: Token, Markov chain

😊😊😠😠😊😊😠😊😠😊😊😊😠😊😊
😠😊😠❓


present
state



⅕ = 0.20
⅘ = 0.80
present
state

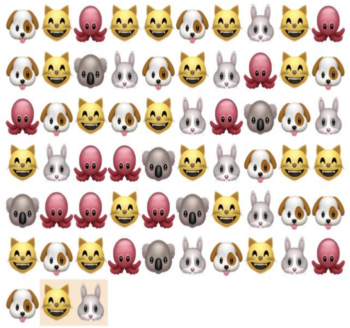
Activity 2. N-imals: Introducing N-grams

Activity 2

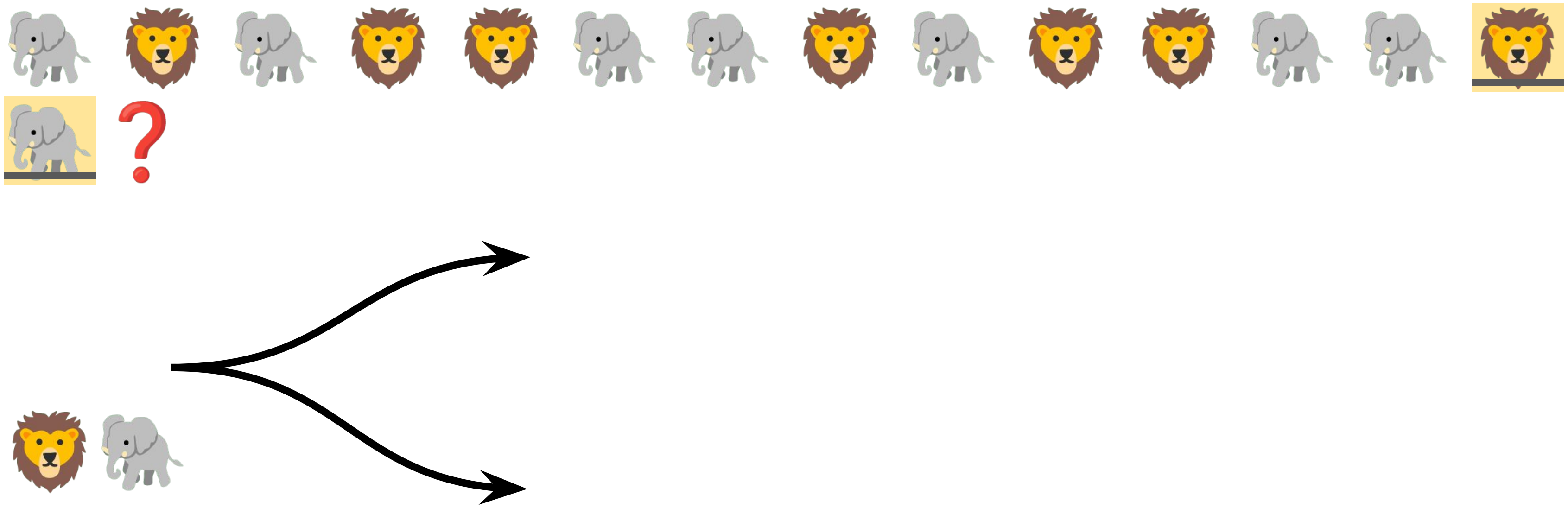
N-imals: Introducing N-grams

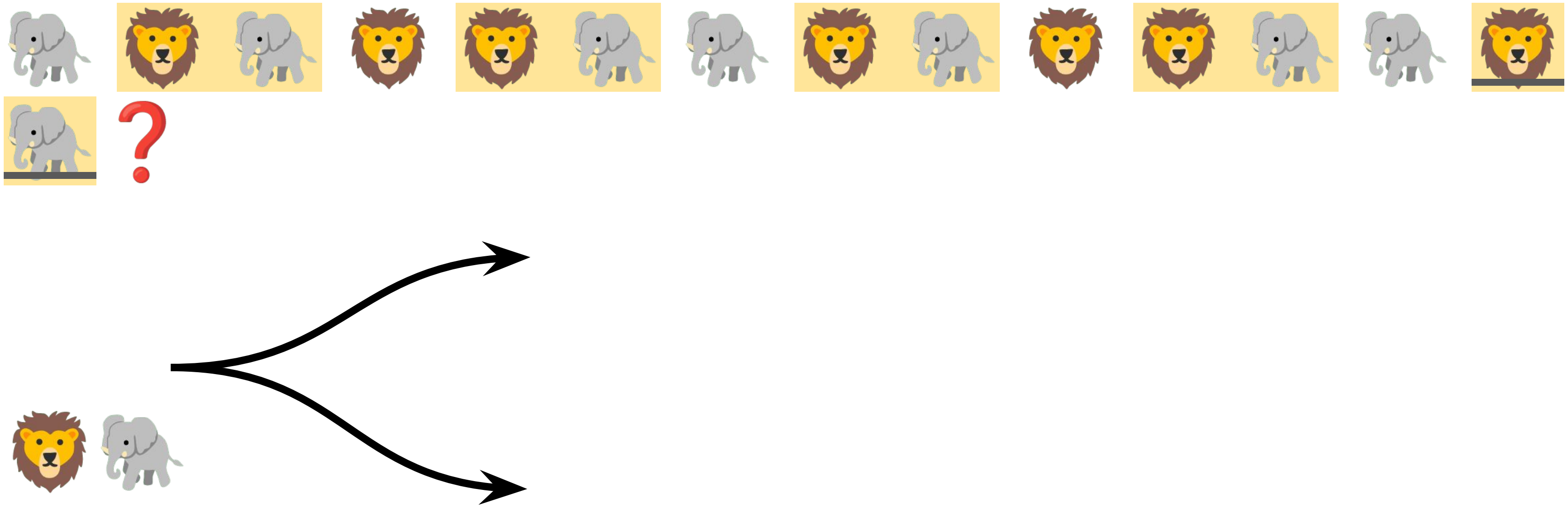
Using simple sequences of animal emojis, learners will predict the following animal emoji following the last n-number of tokens

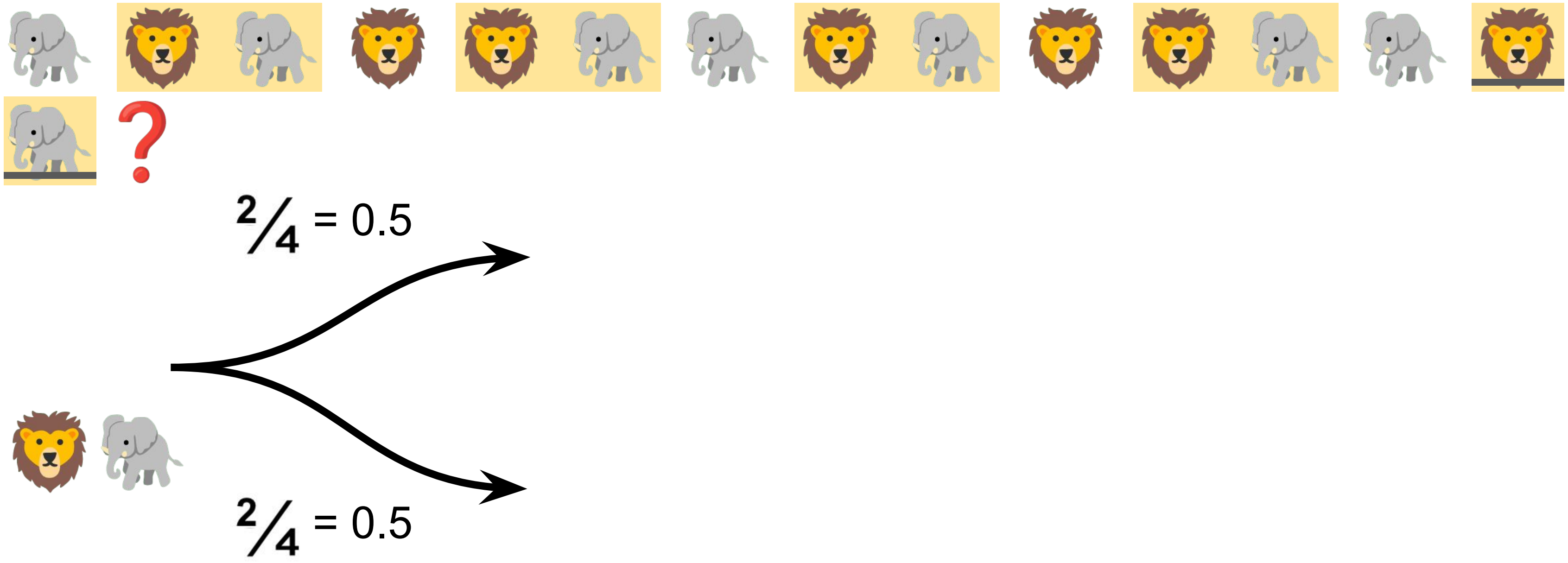
Key Concepts: More context can improve predictions
Key Terms: N-gram, Bigram, Trigram



Answer Key		
😺🐰 → 😺	1/7	= 0.14
😺🐰 → 🐰	0/7	= 0
😺🐰 → 🐙	4/7	= 0.57
😺🐰 → 🐨	0/7	= 0
😺🐰 → 🐶	2/7	= 0.29



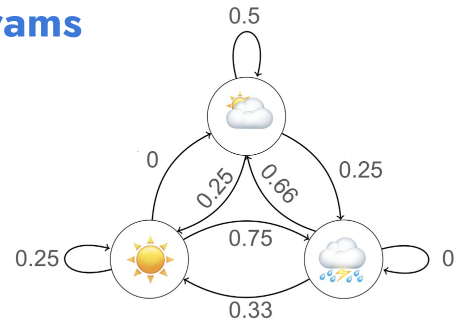



2/4 = 0.5
2/4 = 0.5

Activity 3. Sunny, Rainy, Cloudy

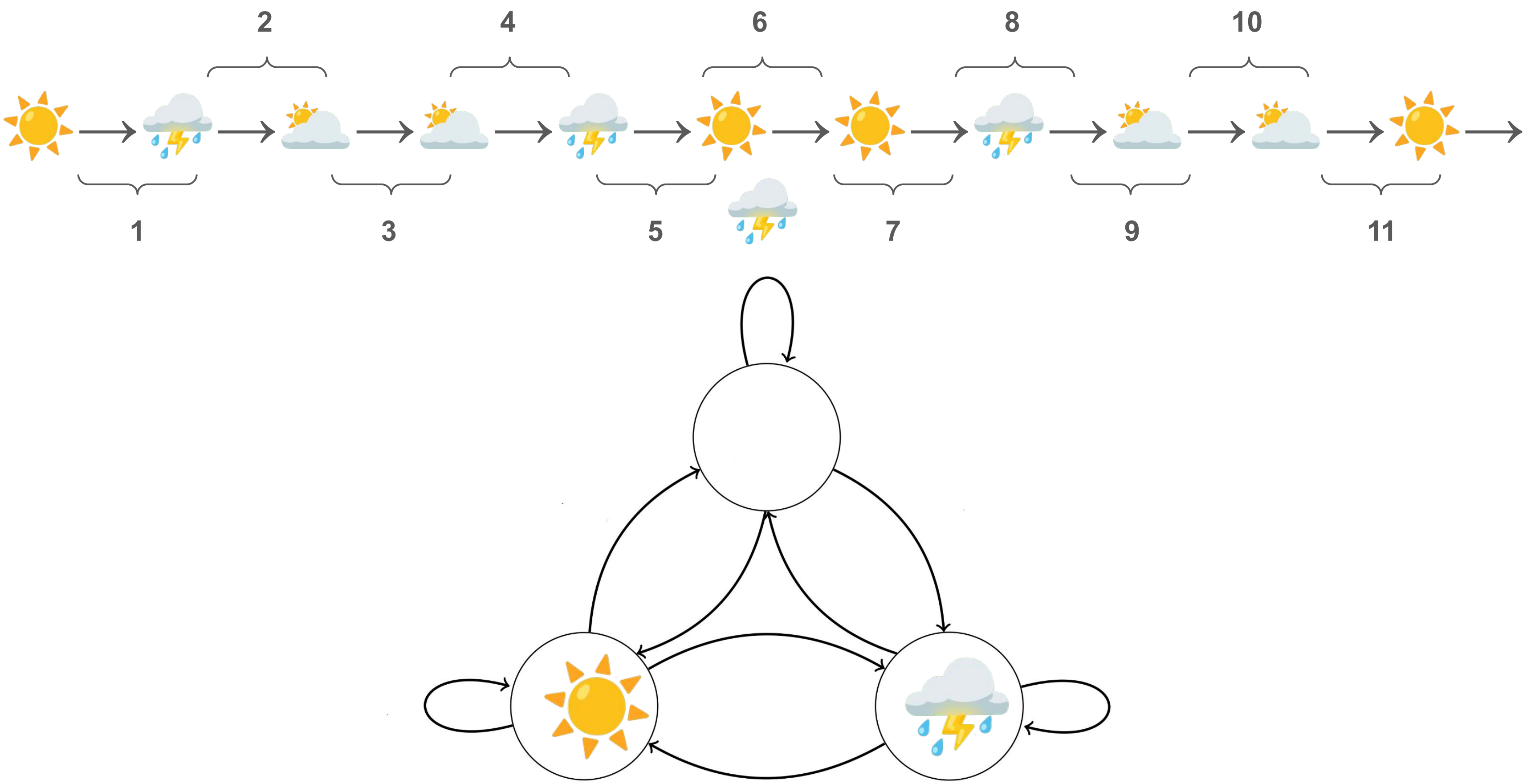
Activity 3

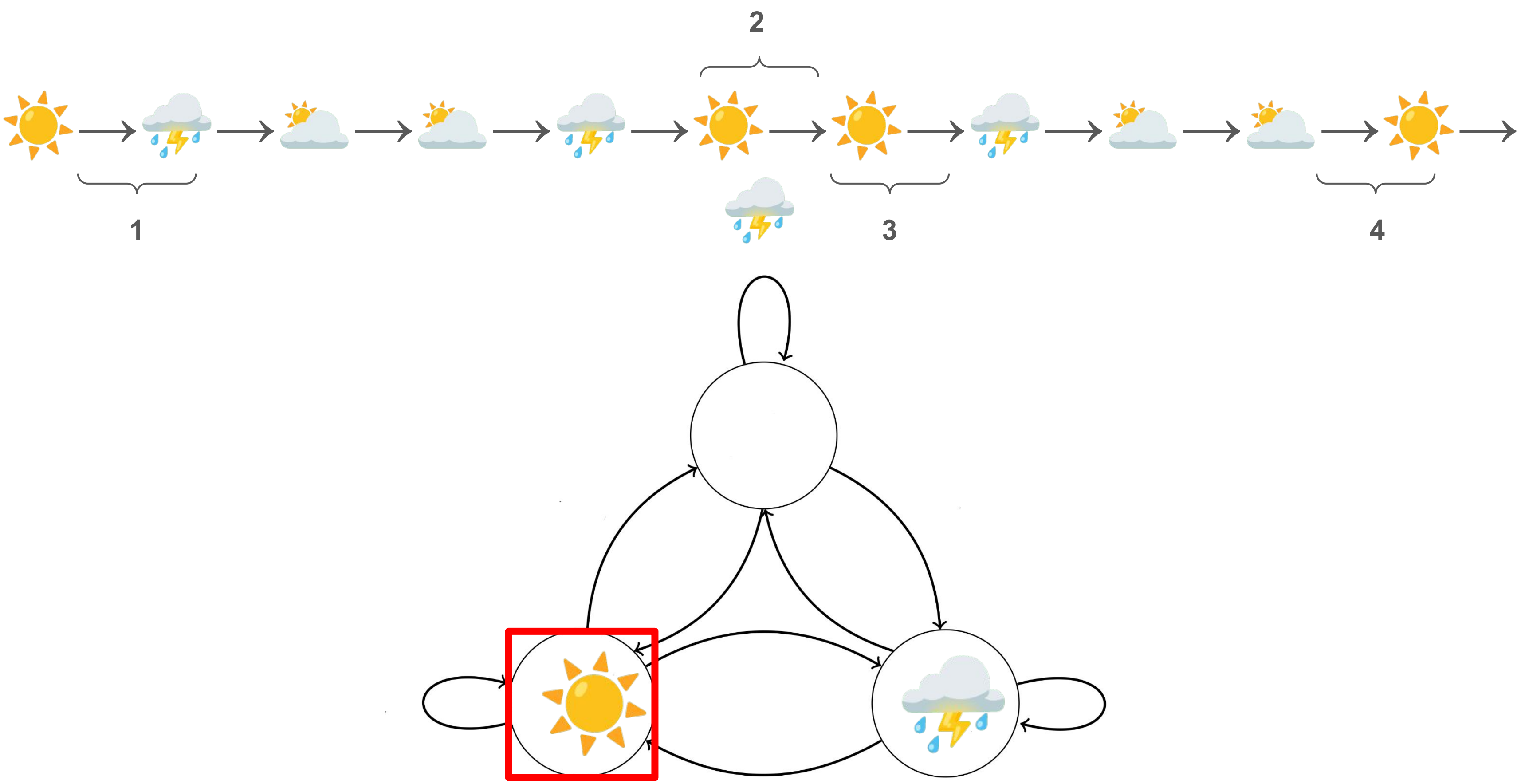
Sunny, Rainy, Cloudy: Introducing State Diagrams

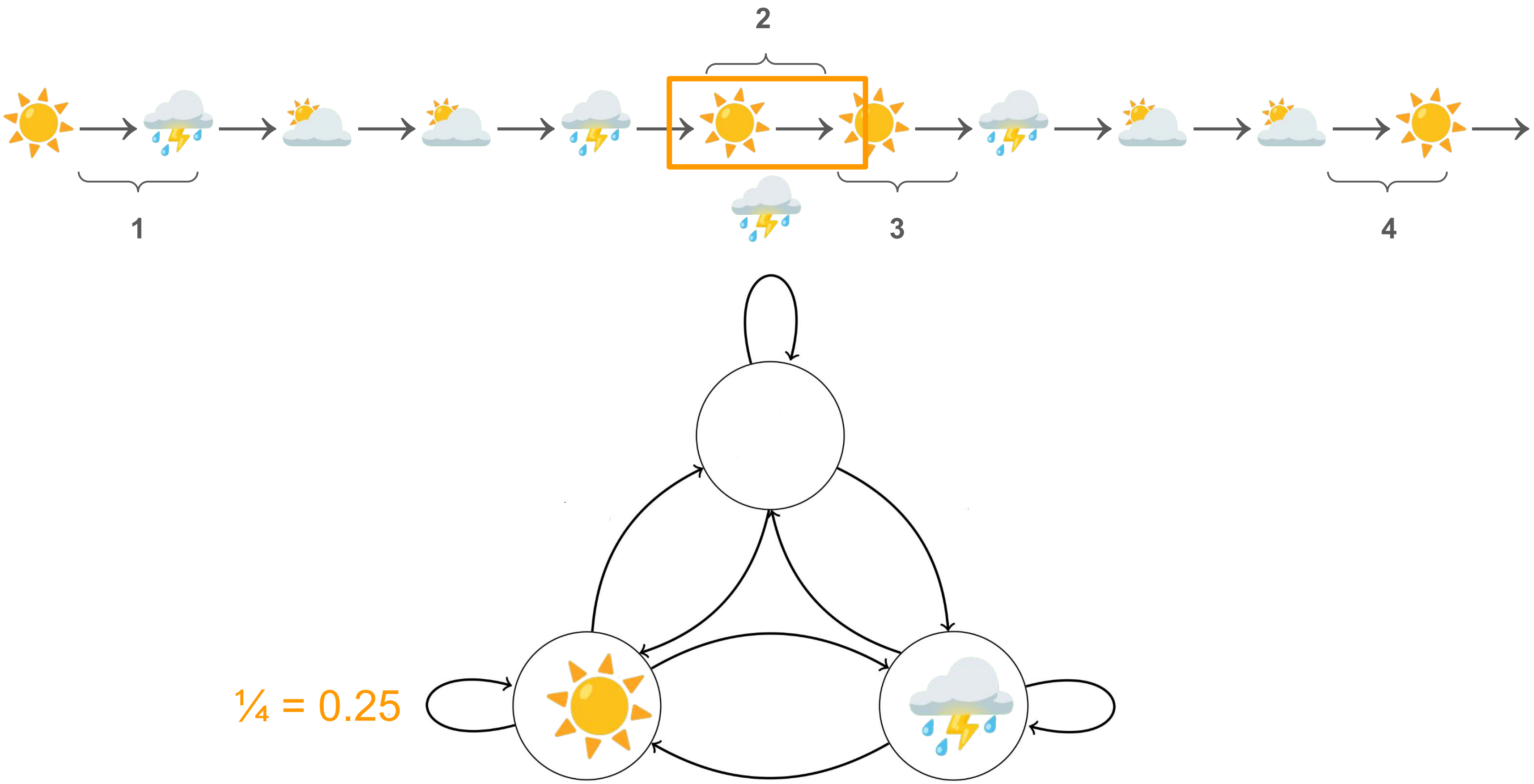


Using a sequence of weather data, learners will create a state diagram and then use it to predict (most likely) future weather.

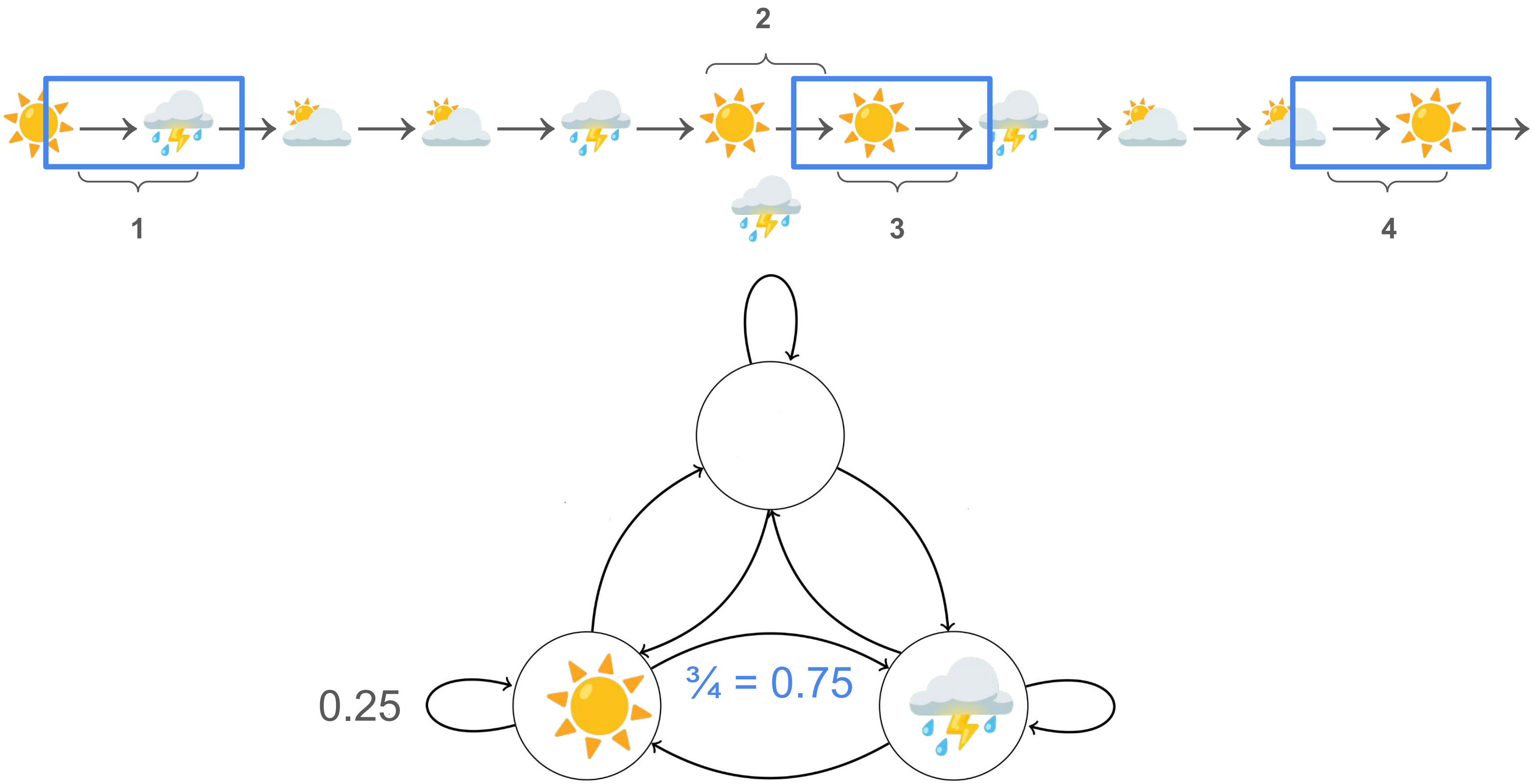
Key Concepts: State transition diagrams, prediction models as a system
Key Terms: State, state diagram, transition, model

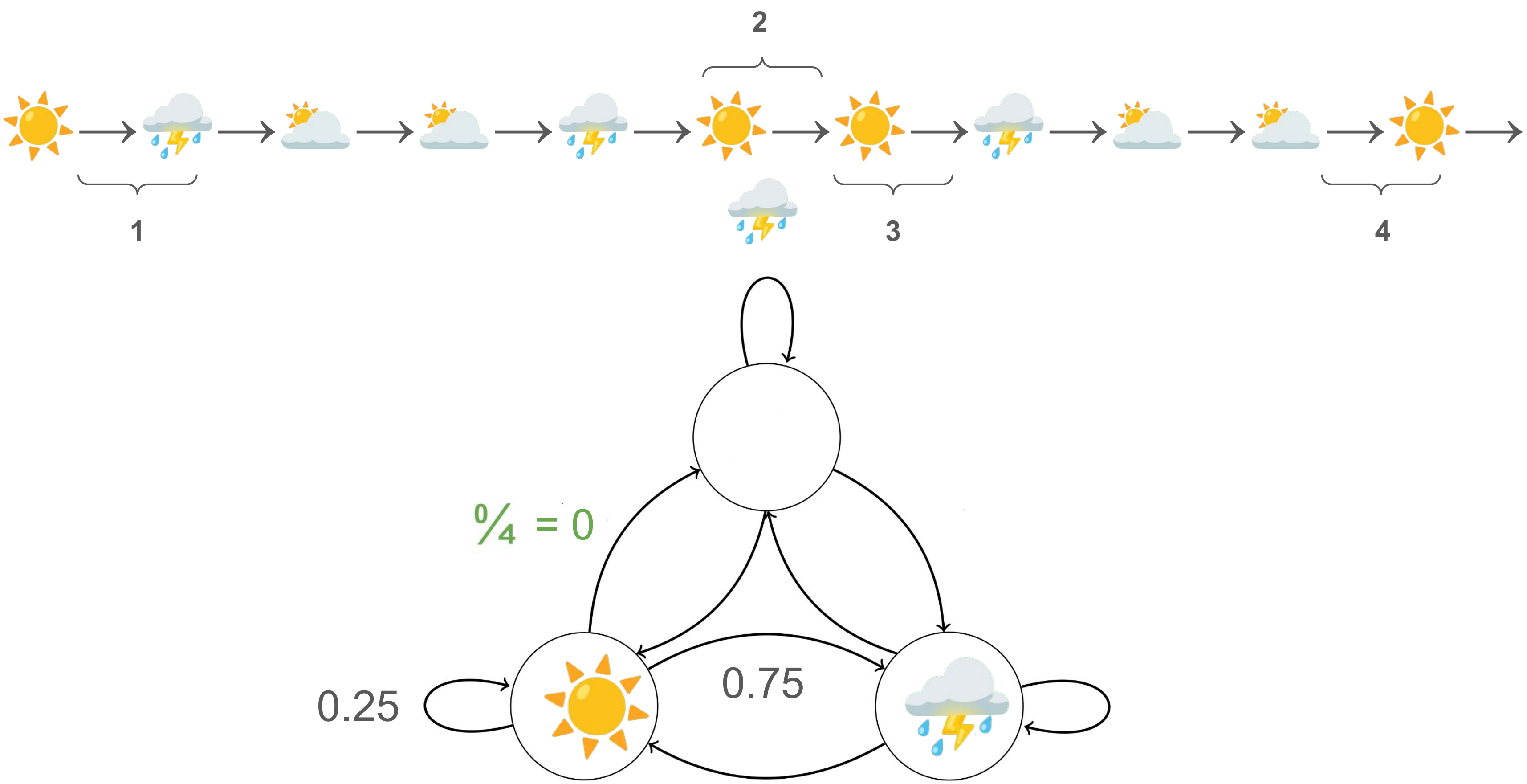

2
4
6
8
10
1
3
5
7
9
11


2
1
3
4

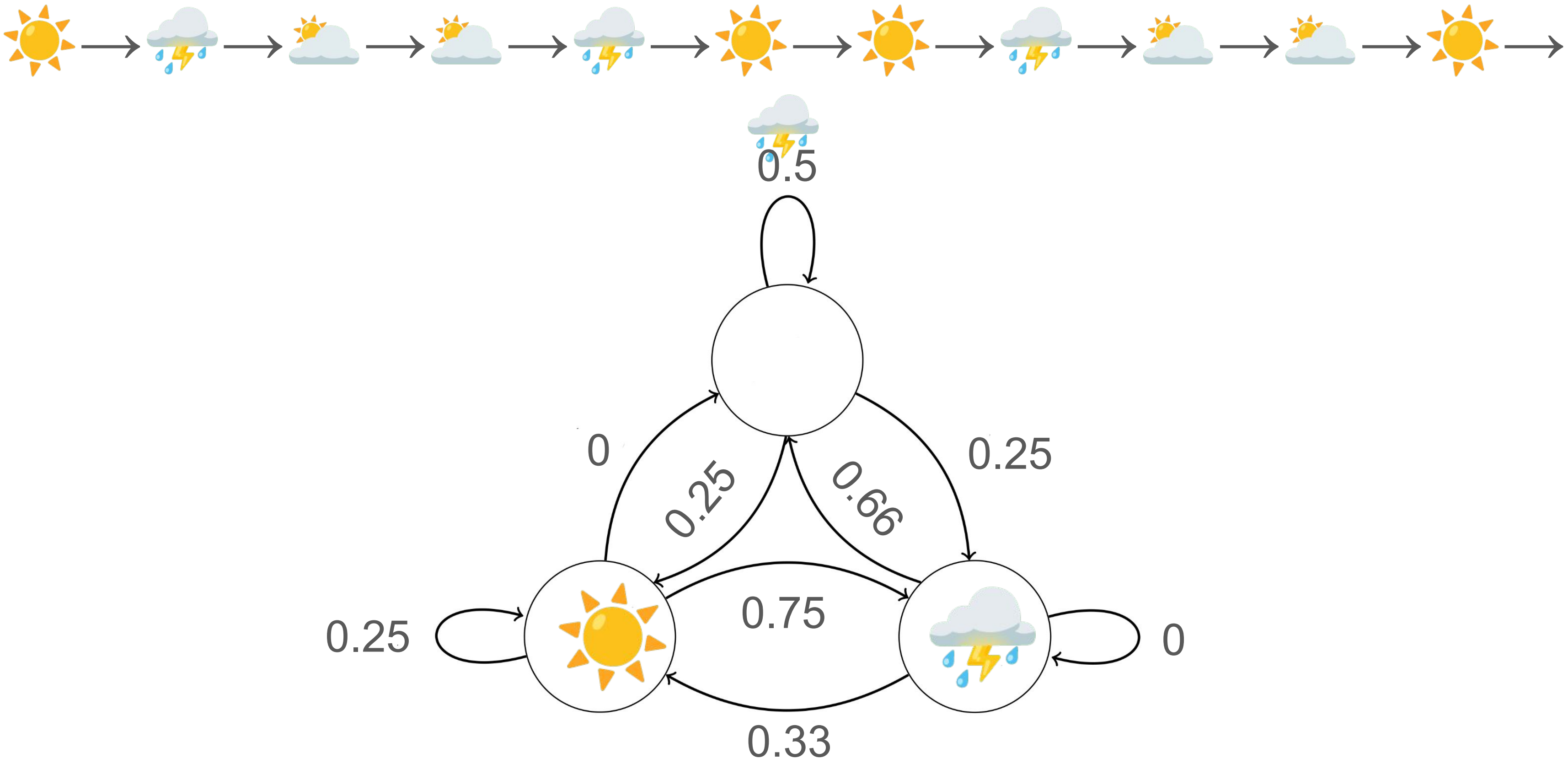


2
1
3
4
¼ = 0.25


2
1
3
4
¾ = 0.75
0.25



1
2
3
4
⁰⁄₄ = 0
0.25
0.75



0.5
0
0.25
0.25
0.66
0.25
0.75
0
0.33

Activity 4. Rock, Paper, Scissors

Activity 4 **Rock, Paper, Scissors: Rapid Prototyping of Datasets**

Using the familiar game of Rock, Paper, Scissors, learners will iteratively build their own datasets, resulting in a final round determined by the output of their dataset.

👊 ✌️ ✋ →
❓

Key Concepts: Data quantity and quality, bias, overfitting
Key Terms: Dataset, data bias, model fit (over/under)

✊ ✋ ✌️

https://danieljnoh.com/rps-markov/

Player 1

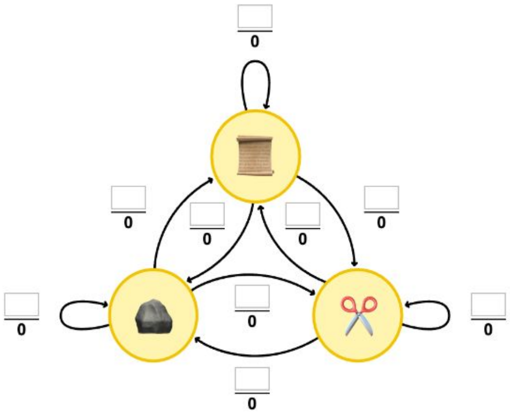
Undo

Next likely move:

?

Fill all 9 numerators to reveal

Dataset



Player 2

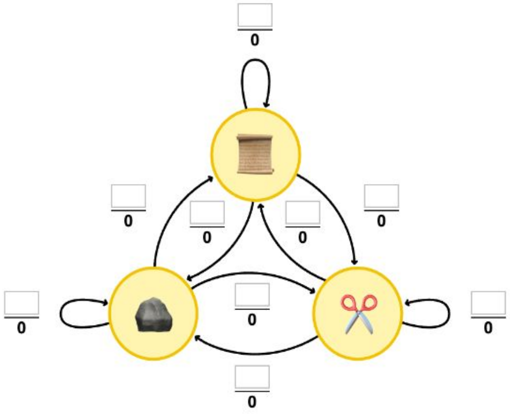
Undo

Next likely move:

?

Fill all 9 numerators to reveal

Dataset



Questions?